Assume $h \geq 2$ and $p \geq 0$. Choose $\vec{d}$ and $\vec{e}$ s.t. $\vec{d} - \vec{e} = (\overset{\text{1 source}}{1}, 0, \dots, 0, \underset{\text{1 sink}}{-1})$. $\therefore \vec{g} = (\underbrace{1, 1, 1, \dots, 1}_{h})$.

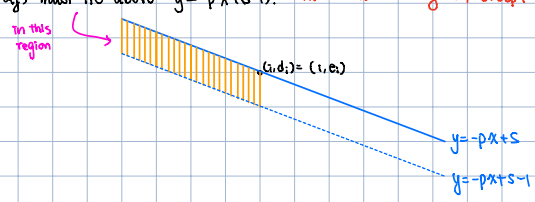
* If $h=1$ or $p \leq 0$ ($\Rightarrow d_i = e_i \leq e_h \ \forall\, 0 \leq i \leq h$) there is at most one nest.

Let $y = -px + s$ be the highest line of slope $-p$ that passes through one of the heads (= feet in this case) $(i, d_i) = (i, e_i)$ for some $0 < i < h$.

Suppose (i, d_i) is on this highest line. Then every point (j, d_j) must lie weakly below the line (j, d_j) $\forall\, 0 < j < h$.

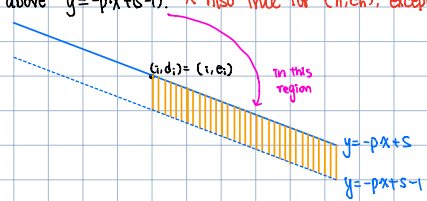
Recall that for heads P, Q (except the head (h, d_h)), there exists a line with slope $-p$ passing above both P and Q but below $P + (0,1)$:

For $0 < j < i$, (Choose $P = (j, d_j)$ and $Q = (i, d_i)$) we have
(j, d_j) must lie above $y = -px + (s-1)$. * Also true when $j = 0$, except $(0, d_0)$ can lie above $y = -px + s$.

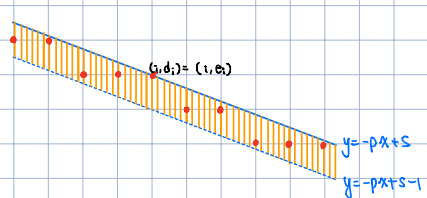


Recall that for feet P, Q (except the feet $(0, e_0)$), there exists a line with slope $-p$ passing above both P and Q but below $Q + (0,1)$:

For $i < j < h$, (Choose $P = (i, e_i)$ and $Q = (j, e_j)$) we have
(j, d_j) must lie above $y = -px + (s-1)$. * Also true for (h, e_h), except (h, e_h) can lie above the line $y = -px + s$.



$\therefore$ All heads (= feet) $(j, d_j) = (j, e_j)$ $\forall\, 1 \leq j \leq h-1$ lie between $y = -px + s$ and $y = -px + s - 1$.
i.e. They are the highest lattice point below $y = -px + s$.



WLOG, we can assume $e_h = 0$ (shift the diagram vertically) so that the sink $(h, e_h) = (h, 0)$ is on the x-axis.

Case I) $(h, 0)$ is below $y = -px + s$ (i.e. $h < \frac{s}{p}$)
Then $(h, 1)$ is above $y = -px + s$ (b/c $(h, 0)$ is above $y = -px + s - 1$) $\Rightarrow$ every lattice path below $y = -px + s$ that ends at $(h, 0)$ must end with an E-step.

Case II) $(h, 0)$ is above $y = -px + s$ (i.e. $h > \frac{s}{p}$)
If π is a nest in the den, then it is an E-end path with the last step $(h-1, 0) \to (h, 0)$. Since $(h-1, 0)$ is a lattice point on $x = h-1$, it lies below $y = -px + s$. Hence deleting the last E-step gives a lattice path from $(0, d_0)$ to $(h-1, 0)$ below $y = -px + s$.
hence $h - 1 < \frac{s}{p}$

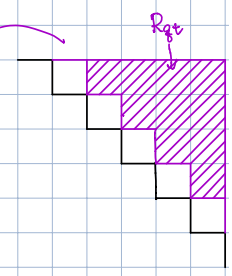
$\therefore$ There is a one-to-one correspondence :

Note that in both cases, the end-point $(\ell, 0)$ satisfies $\ell \le \lfloor \frac{s}{p} \rfloor$

$\{\pi : \pi \text{ is a nest in the given den}\} \longleftrightarrow \{\text{lattice paths from } (0, d_0) \text{ to either } \underbrace{(n-1, 0) \text{ or } (n, 0)}_{\ell = n-1 \text{ or } n} \text{ weakly below } y = -px + s \text{ (except for a possible S-run on the y-axis)}\}$

$d_0 \ge \lfloor s \rfloor$ $(\because d_0 > s-1)$

By Theorem 3.5.1, since $g = (1, 1, \dots, 1)$, $R_g = R_0 = R_+ = \{\varepsilon_i - \varepsilon_j : 1 \le i < j \le n\}$ = set of all positive roots ($|\vec{g}| = n$), $R_{g+} = \{\varepsilon_i - \varepsilon_j : 1 \le i < j \le n,\ j \ne i+1\}$



Note that $\vec{e} = (d_0 - 1, d_1, d_2, \dots, d_{n-1}, d_n + 1)$

Hence the weight of H is $(d_0 - e_1, d_1 - e_2, \dots, d_{n-1} - e_n) = (d_0 - d_1, d_1 - d_2, \dots, d_{n-2} - d_{n-1}, d_{n-1} - d_n - 1)$

(e_1: #S-step on $x=0$ in π^0; e_2: #S-step on $x=1$ in π^0; e_n: #S-step on $x=n-1$ in π^0)

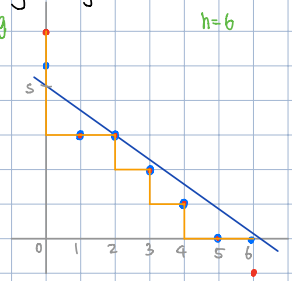
If $(n, 0)$ is above $y = -px + s$, then $(d_0 - d_1, \dots, d_{n-1} - d_n - 1) = (b_1, \dots, b_n) = \vec{b}$ where b_i = #S-steps on $x = i-1$ in Path Theorem 5.5.1 (with $r = \frac{s}{p}$, $\ell = \lfloor \frac{s}{p} \rfloor + 1 = (n-1) + 1 = n$). Hence $H = \mathcal{H}_{\vec{b}}$ in Def. 3.7.1 in path and $\Psi_{\hat{g}}(H) = D_{\vec{b}} \in \mathcal{E}^+$.

notation in path

If $(n, 0)$ is below $y = -px + s$, then H is a "shorter" Catalanimal because $b_n = 0$ (and may be more b_i's). Removing trailing zeros from $\vec{b}$, say λ, we have $\Psi_{\hat{g}}(H) = D_\lambda$. Note that $D_{\vec{b}} \cdot 1 = D_\lambda \cdot 1$ (By Lemma 3.6.2 in path)

Corollary 3.5.2 in this case reduces to the generalized shuffle theorem for paths under $y = -px + s$ including the more general version for paths extended along the y-axis (Remark 5.5.2 in path).

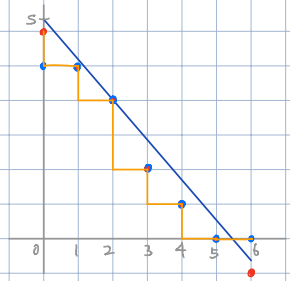
e.g $n = 6$



$(n, 0)$ below $y = -px + s$

$\vec{b} = (3, 0, 1, 1, 1, 0)$

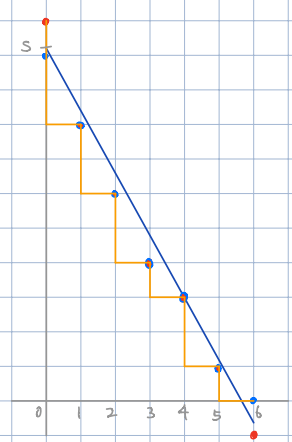
$\lambda = (3, 0, 1, 1, 1)$



$(n, 0)$ above $y = -px + s$

$\vec{b} = (1, 1, 2, 1, 1, 0)$

still possible to have trailing zeros



$(n, 0)$ above $y = -px + s$

$\vec{b} = (3, 2, 2, 1, 2, 1)$

possible to have nonzero b_ℓ